

Olympics Medal Prediction: Reproduction Study

Machine Learning Mini Project (Reproduce and Extend) | Part 1: Reproduction

Team: [Name 1, SRN] and [Name 2, SRN] | Reference: [paper title, authors, year] | Repo: [GitHub link]

1. Problem Statement

Given a country's economic, demographic and Olympic-participation data, predict its performance at the Summer Olympics. We solve two tasks, following the reference: (i) **classification**, whether a country wins at least one medal (metric: accuracy), and (ii) **regression**, how many medals it wins (metric: RMSE). The goal of this part is to reproduce the reference's reported results and explain any differences.

2. Dataset

- **Olympic data:** Summer Games 1988-2016, medals and distinct athletes per country per Games (Kaggle *athlete_events.csv*). Medals are counted once per event, not per athlete.
- **Economic and demographic data:** World Bank WDI via the *wbgapi* package: GDP, GDP per capita, GDP growth, population, population growth, land area. Olympic country codes were mapped to World Bank ISO codes.
- **Derived features:** population share of world, athletes share of Games, medals at the previous Games, total medals awarded that year. Rows with missing values are dropped, as in the reference.
- **Features used (9):** Year, GDP_Growth, GDP_Per_Capita, Pop, Pop_pct_World, Athletes, Athletes_pct, Medals_Last_Games, Total_Medals_Year.
- **Split:** train on 1988-2008, validate on 2012 (197 countries), final test on 2016 (197 countries). For the final test, training uses 1988-2012.

3. Approach and Implementation

- **Preprocessing:** every model except Random Forest sits in a pipeline with *StandardScaler*. Features differ by orders of magnitude (Pop $\sim 1e9$, GDP_Growth ~ 1); without scaling, SVM, Naive Bayes and MLP scored only 0.58-0.61 accuracy.
- **Classification models:** Logistic Regression, SVM (polynomial kernel, degree 1), Gaussian Naive Bayes, MLP, Random Forest. Hyperparameters tuned by 3-fold cross-validation on training years only (grid search or randomized search).
- **Regression models:** Linear, Ridge, Lasso, SVR (linear), Poisson, Random Forest, and a weighted Linear Regression. Each is trained on all training rows and evaluated on raw predictions, with no classifier gate.
- **Final model (2016):** Gaussian Naive Bayes predicts whether a country wins any medal; Ridge regression predicts the count for countries classified as medal winners (others get 0, negatives clipped to 0).
- **Code:** Python 3.11, scikit-learn, pandas, numpy; *random_state=42* throughout. A single script (*src/02_run_reference_models.py*) runs all models and writes the comparison tables.

4. Results: Reported vs Our Result

Model (2012 classification)	Reported acc.	Our acc.	Diff
Logistic Regression	0.8832	0.8832	0.0000
SVM	0.8629	0.8680	+0.0051
Gaussian Naive Bayes	0.7614	0.8883	+0.1269
MLP	0.8782	0.8731	-0.0051
Random Forest	0.8731	0.8731	0.0000

Model (2012 regression)	Reported RMSE	Our RMSE	Diff
Linear Regression	3.1115	3.2068	+0.0953
Ridge	3.1582	3.1868	+0.0286
Lasso	3.0386	3.2515	+0.2128
SVR	2.8156	3.3439	+0.5283
Poisson	5.8460	5.5025	-0.3435
Random Forest	2.8869	2.5457	-0.3411
Weighted Linear Regression	6.6632	4.3064	-2.3568

Final model (2016 test)	Reported RMSE	Our RMSE	Diff
Gaussian NB + Ridge	4.8243	3.9663	-0.8579

5. Discussion of Differences

Logistic Regression, SVM, MLP and Random Forest are within 0.5 percentage points of the reported accuracy. Linear, Ridge and Lasso regression are within about 0.2 RMSE. Two results differ clearly:

- **Gaussian Naive Bayes** (0.888 vs 0.761) and **Weighted Linear Regression** (4.31 vs 6.66). Naive Bayes is very sensitive to scaling and variance smoothing, and the weighted model depends on a weighting function that the reference does not fully specify. We could not identify the exact settings.
- **SVR** (3.34 vs 2.82) and the **2016 final RMSE** (3.97 vs 4.82) also differ. Our final model flags more countries as medal winners (36%) than the reference output does (28%), which changes which countries receive a regression prediction.

Likely causes: our merged dataset was rebuilt from public sources and may differ slightly from the reference's (row counts match for 2012 and 2016, but feature values may not); unreported preprocessing and hyperparameter search ranges; random seeds; and scikit-learn version differences. Our linear baseline on the same split is 3.21 vs the reference 3.11, which suggests part of the gap comes from the data itself.

6. Conclusions

- The reference pipeline is reproducible in structure: feature scaling was the decisive fix, raising SVM, Naive Bayes and MLP accuracy from about 0.6 to 0.87-0.89.
- Classification results reproduce closely for four of five models. Regression results reproduce approximately, with Random Forest and linear models giving the lowest 2012 RMSE (about 2.5-3.3) and Poisson and the weighted model the worst.
- Our lower 2016 RMSE (3.97 vs 4.82) is a difference from the reference, not an improvement: we cannot rule out that it comes from dataset differences rather than the method.
- Remaining work: Part 2 (one additional model on the same split and metrics), if included in the submission.